

Chapter 4

Rational Functions

This week takes the polynomial ideas from Week 3 and adds division. A rational function is a ratio of two polynomials,

$$f(x) = \frac{p(x)}{q(x)}.$$

Because both the numerator and denominator can have roots, rational functions have more visible features than polynomials. Some roots pull the graph down to the x -axis. Other roots make the formula undefined and force the graph toward a vertical line.

The graph is governed by the interplay between numerator and denominator. Where the numerator p is zero, the function is zero. Where the denominator q is zero, the function is undefined and the graph can run off toward a vertical asymptote. The relative sizes of the two degrees decide what happens at the far ends. As with polynomials, factored form displays the local features, and leading terms settle the end behavior.

4.1 Zeros and Vertical Asymptotes

The first features to locate are zeros and vertical asymptotes. Both come from roots, but they come from different parts of the fraction.

The numerator and denominator each contribute their own roots, and the two kinds behave oppositely. A root of the numerator (that is not also a root of the denominator) makes the whole fraction zero, so the graph crosses or touches the axis there exactly as a polynomial would. A root of the denominator makes the fraction undefined; near such a point the denominator is tiny while the numerator is not, so the quotient grows without bound. The graph approaches a vertical line called a vertical asymptote.

Definition: Zero and vertical asymptote. For $f(x) = p(x)/q(x)$ in lowest terms, a *zero* is an input where $p(x) = 0$, producing $f(x) = 0$. A *vertical asymptote* is a vertical line $x = a$ where $q(a) = 0$; as $x \rightarrow a$ the values of f increase or decrease without bound.

Key idea. Numerator roots pull the graph *to* the axis (zeros). Denominator roots push the graph *away* along a vertical line (asymptotes). The factored form of p/q lists both at once.

The picture. At a numerator root the fraction reads (small)/(ordinary), which is small: the graph sits near the x -axis. At a denominator root the fraction reads (ordinary)/(small), which is enormous: the graph races up or down beside the vertical line $x = a$. The two roots are mirror situations, one driving the value to zero, the other to infinity.

Example 4.1.1. (*locating zeros and asymptotes*) For

$$f(x) = \frac{(x-2)(x+1)}{(x-3)(x+1)}$$

the numerator is zero at $x = 2$ and $x = -1$; the denominator is zero at $x = 3$ and $x = -1$. The factor $(x+1)$ is common to both, so it cancels: it is neither a true zero nor an asymptote, but a single missing point (a hole) at $x = -1$. After cancellation the function is $f(x) = \frac{x-2}{x-3}$ with a genuine zero at $x = 2$ and a vertical asymptote at $x = 3$.

The common factor is important because it tells us that the original formula was undefined at $x = -1$, even though the simplified formula has the same values elsewhere. Graphically, that leaves a missing point rather than a whole vertical asymptote.

Common pitfall. A factor common to numerator and denominator does not create an asymptote. It cancels, leaving a hole, a single point removed from the graph. Always reduce to lowest terms before declaring zeros and asymptotes; the leftover denominator roots are the asymptotes, the leftover numerator roots are the zeros.

4.1.1 Sign analysis across the features

Once zeros and vertical asymptotes are marked, the graph is broken into intervals. On each interval the function is continuous and keeps one sign. This means the same sign-chart idea from polynomials still works, but now the feature points include both zeros and vertical asymptotes.

Zeros and vertical asymptotes are the only places a rational function in lowest terms can change sign, for the same reason as with polynomials: the value can switch sign only by passing through zero (a numerator root) or through infinity (a denominator root). Marking both kinds on a number line and testing one point per interval gives the sign on each interval, just as before.

Example 4.1.2. (*sign chart for a rational function*) Take $f(x) = \frac{x-2}{x-3}$. The feature points are the zero $x = 2$ and the asymptote $x = 3$.

Interval	test x	$x - 2$	$x - 3$	sign of f
$(-\infty, 2)$	0	−	−	+
$(2, 3)$	2.5	+	−	−
$(3, \infty)$	4	+	+	+

The graph is above the axis left of 2, dips below between 2 and 3, and returns above the axis past the asymptote at 3.

This chart gives the side of the x -axis but not the exact height of the graph. Near $x = 3$, the asymptote supplies the missing information: the curve must escape upward or downward instead of passing through the vertical line.

4.2 End Behavior: Comparing Degrees

Zeros and vertical asymptotes describe what happens near special finite inputs. End behavior asks a different question: what does the graph do far away from the origin? For rational functions, the answer comes from comparing the size of the numerator to the size of the denominator.

Far from the origin the leading terms of numerator and denominator dominate all others, so the end behavior of $p(x)/q(x)$ matches that of the ratio of leading terms. Writing $p(x) = a_m x^m + \cdots$ and $q(x) = b_n x^n + \cdots$, the comparison of the degrees m and n decides the outcome.

Key idea. End behavior of $f(x) = \frac{a_m x^m + \cdots}{b_n x^n + \cdots}$

- $m < n$ (*proper, numerator smaller*). The ends approach the horizontal asymptote $y = 0$.
- $m = n$ (*equal degrees*). The ends approach the horizontal asymptote $y = \frac{a_m}{b_n}$, the ratio of leading coefficients.
- $m > n$ (*improper, numerator larger*). There is no horizontal asymptote. Polynomial long division produces a polynomial part the ends follow; when $m = n+1$ that part is a line, a *slant asymptote*.

Example 4.2.1. (*horizontal asymptote from equal degrees*) For $f(x) = \frac{3x^2 - 1}{2x^2 + x + 5}$ the degrees are equal at 2. The ends approach $y = \frac{3}{2}$, the ratio of the leading coefficients 3 and 2. The lower-degree terms affect the middle of the graph but vanish in the comparison at the ends.

The horizontal asymptote is not found by plugging in a convenient value of x . It is a statement about what the graph approaches when x moves far to the left or far to the right.

Example 4.2.2. (*horizontal asymptote at zero*) For $f(x) = \frac{4x+1}{x^2-9}$ the numerator degree 1 is below the denominator degree 2, so the ends approach $y = 0$. The denominator factors as $(x-3)(x+3)$, giving vertical asymptotes at $x = 3$ and $x = -3$; the single numerator root $x = -\frac{1}{4}$ is the only zero.

4.3 Improper Rational Functions and Long Division

When the numerator degree is smaller than the denominator degree, the fraction itself shrinks toward zero at the ends. When the degrees are equal, the ratio of leading coefficients gives a horizontal asymptote. The remaining case is different: the numerator is at least as large in degree as the denominator.

When the numerator degree is at least the denominator degree, the function is *improper*, and the honest description of its ends comes from polynomial long division. Dividing p by q rewrites the function as a polynomial quotient plus a proper remainder term:

$$\frac{p(x)}{q(x)} = Q(x) + \frac{R(x)}{q(x)}, \quad \deg R < \deg q.$$

Far from the origin the remainder term $R(x)/q(x)$ is proper, so it decays to zero, and the graph approaches the polynomial $Q(x)$. The quotient Q is the end-behavior curve. This is the rational-function version of listening to the dominant part of the formula at large scale.

Key idea. For an improper rational function, long division gives $f(x) = Q(x) + \frac{R(x)}{q(x)}$ with the remainder term proper. The ends of the graph hug the quotient $Q(x)$, because the proper part dies off. When $\deg p = \deg q + 1$ the quotient is a line, and that line is the slant asymptote.

Example 4.3.1. (*a slant asymptote by long division*) Analyze $f(x) = \frac{x^2 + x - 2}{x - 3}$.

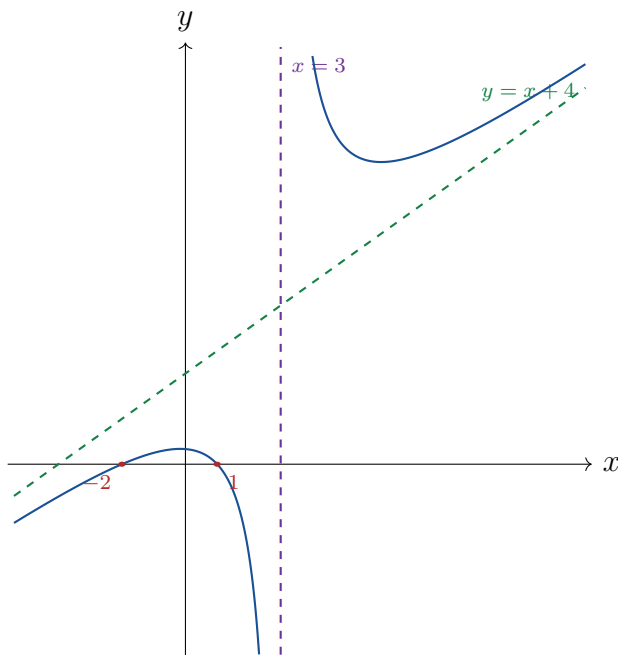
The numerator degree 2 exceeds the denominator degree 1 by one, so expect a slant asymptote. The division will separate the function into a line plus a leftover fraction that becomes small at the ends. Divide $x^2 + x - 2$ by $x - 3$:

$$\begin{array}{r} x+4 \\ x-3 \overline{) x^2+x-2} \\ \underline{x^2-3x} \\ 4x-2 \\ \underline{4x-12} \\ 10 \end{array}$$

So

$$f(x) = x + 4 + \frac{10}{x - 3}.$$

The remainder term $\frac{10}{x - 3}$ is proper and decays to zero at the ends, so the graph approaches the line $y = x + 4$. That line is the slant asymptote. The factored numerator $x^2 + x - 2 = (x + 2)(x - 1)$ gives zeros at $x = -2$ and $x = 1$, and the denominator gives a vertical asymptote at $x = 3$.



Common pitfall. A slant asymptote appears only when the numerator degree is exactly one more than the denominator degree. If it exceeds by two or more, the quotient $Q(x)$ is a parabola or higher curve, and the ends follow that curve, not a line. There is no horizontal asymptote in any improper case.

Example 4.3.2. (*improper with equal-after-division check*) For $f(x) = \frac{2x^3}{x^2 + 1}$ the numerator degree exceeds the denominator by one. Long division gives

$$\frac{2x^3}{x^2 + 1} = 2x - \frac{2x}{x^2 + 1},$$

so the slant asymptote is $y = 2x$ and the proper remainder $-\frac{2x}{x^2 + 1}$ decays at the ends. The only zero is $x = 0$, and the denominator $x^2 + 1$ is never zero, so there are no vertical asymptotes.

This example is useful because it separates two ideas that students often connect too strongly. A rational function can have a slant asymptote without having any vertical asymptotes. Vertical asymptotes come from real roots of the denominator; slant asymptotes come from degree comparison and division.

4.4 Assembling the Graph

A rational graph can look complicated because different pieces of information are happening at once: zeros, holes, vertical asymptotes, end behavior, and signs. The way to make the sketch manageable is to read these features one at a time.

A faithful sketch of a rational function in lowest terms combines several readings, each taken from the factored form or the long division.

Key idea. Sketching a rational function.

1. *Reduce* to lowest terms; note any holes from cancelled factors.
2. *Zeros*: numerator roots, with crossing or touching set by multiplicity.
3. *Vertical asymptotes*: denominator roots; the graph escapes to $\pm\infty$ beside each.
4. *End behavior*: compare degrees for a horizontal asymptote, or divide for a slant or polynomial asymptote.
5. Use a sign chart over zeros and asymptotes to keep the curve on the correct side of the axis between features.

Example 4.4.1. (*a complete analysis*) Analyze $f(x) = \frac{x^2 - 1}{x^2 - 4}$.

We will read the function in the standard order: reduce first, then locate zeros and vertical asymptotes, then determine end behavior, and finally use signs to guide the sketch.

Reduce. Numerator $(x - 1)(x + 1)$, denominator $(x - 2)(x + 2)$; no common factors, so no holes.

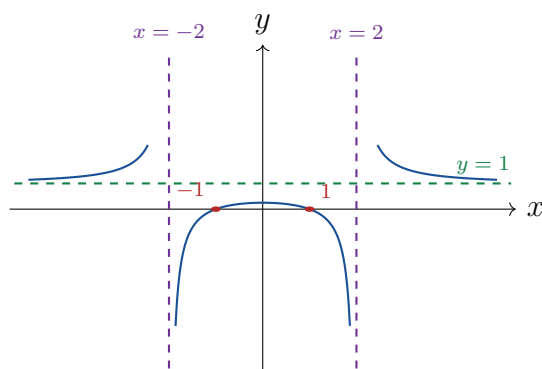
Zeros. $x = 1$ and $x = -1$, both simple, so the graph crosses there.

Vertical asymptotes. $x = 2$ and $x = -2$.

End behavior. Equal degrees, leading coefficients both 1, so the horizontal asymptote is $y = 1$.

Sign chart. Feature points $-2, -1, 1, 2$ split the line into five intervals; testing one point in each places the curve above the axis on the outer intervals (approaching $y = 1$), with the expected escapes beside $x = \pm 2$.

The sketch is not produced by guessing the curve all at once. It is assembled from features. The asymptotes form a frame, the zeros pin the curve to the axis, the horizontal asymptote controls the outer branches, and the sign information keeps each piece on the correct side of the axis.



Interactive. Build a rational function by placing numerator zeros and vertical asymptotes, each with a multiplicity, in factored form, then fit a hidden target. The fit score weighs zeros, asymptotes, multiplicities, and overall agreement, and the current degree ratio reports the end behavior. Compare how a numerator root drives the curve to the axis while a denominator root drives it to a vertical line.

https://bobovski66.github.io/Precalculus/demos/rational_functions.html

Use the lab to practice reading the graph as a set of features rather than as one large drawing problem. Move one zero or one asymptote at a time and watch which part of the graph responds.

4.5 Exercises

Exercises 4.5

4.5.1. For $f(x) = \frac{(x-4)(x+2)}{(x-1)(x+5)}$, state the zeros and the vertical asymptotes, and confirm there are no common factors.

4.5.2. Identify the hole, the zeros, and the vertical asymptotes after reducing to lowest terms:

$$f(x) = \frac{(x-3)(x+2)}{(x+2)(x-5)}.$$

4.5.3. Explain, in terms of (small)/(ordinary) versus (ordinary)/(small), why a numerator root produces a zero while a denominator root produces a vertical asymptote.

4.5.4. Give the horizontal asymptote, if any, of each: (a) $\frac{2x+1}{x-7}$; (b) $\frac{5x^2-3}{x^2+4}$; (c) $\frac{x+6}{x^3-1}$; (d) $\frac{x^3}{x+2}$.

- 4.5.5. For $f(x) = \frac{6x^2 + x}{3x^2 - 12}$, find the horizontal asymptote from the leading coefficients, the zeros, and the vertical asymptotes.
- 4.5.6. Use long division to write $f(x) = \frac{x^2 - 5x + 6}{x - 1}$ in the form $Q(x) + \frac{R}{x - 1}$, and state the slant asymptote.
- 4.5.7. Find the slant asymptote of $f(x) = \frac{2x^2 + 3x - 1}{x + 2}$ by long division, and give the zeros and the vertical asymptote.
- 4.5.8. Explain why an improper rational function never has a horizontal asymptote, and state the condition under which its end-behavior curve is a straight line.
- 4.5.9. Build a complete sign chart for $f(x) = \frac{x - 1}{(x + 2)(x - 3)}$ over its zeros and vertical asymptotes, recording the sign of each factor in each interval.
- 4.5.10. Carry out a full analysis of $f(x) = \frac{x^2 - 4}{x^2 - 1}$ (reduce, zeros, vertical asymptotes, end behavior, sign chart) and sketch a graph consistent with all of it.
- 4.5.11. A rational function has zeros at $x = 0$ and $x = 3$, vertical asymptotes at $x = -1$ and $x = 2$, and a horizontal asymptote at $y = 1$. Construct a factored formula meeting all four conditions, and explain how you chose the degrees.
- 4.5.12. Open the Zeros and Asymptotes Lab. Build a rational function with at least two vertical asymptotes and one repeated numerator zero, record its factored form, the degree ratio, and the fit score against a target, and write one sentence on how the end behavior changed when you altered the degree of the numerator.